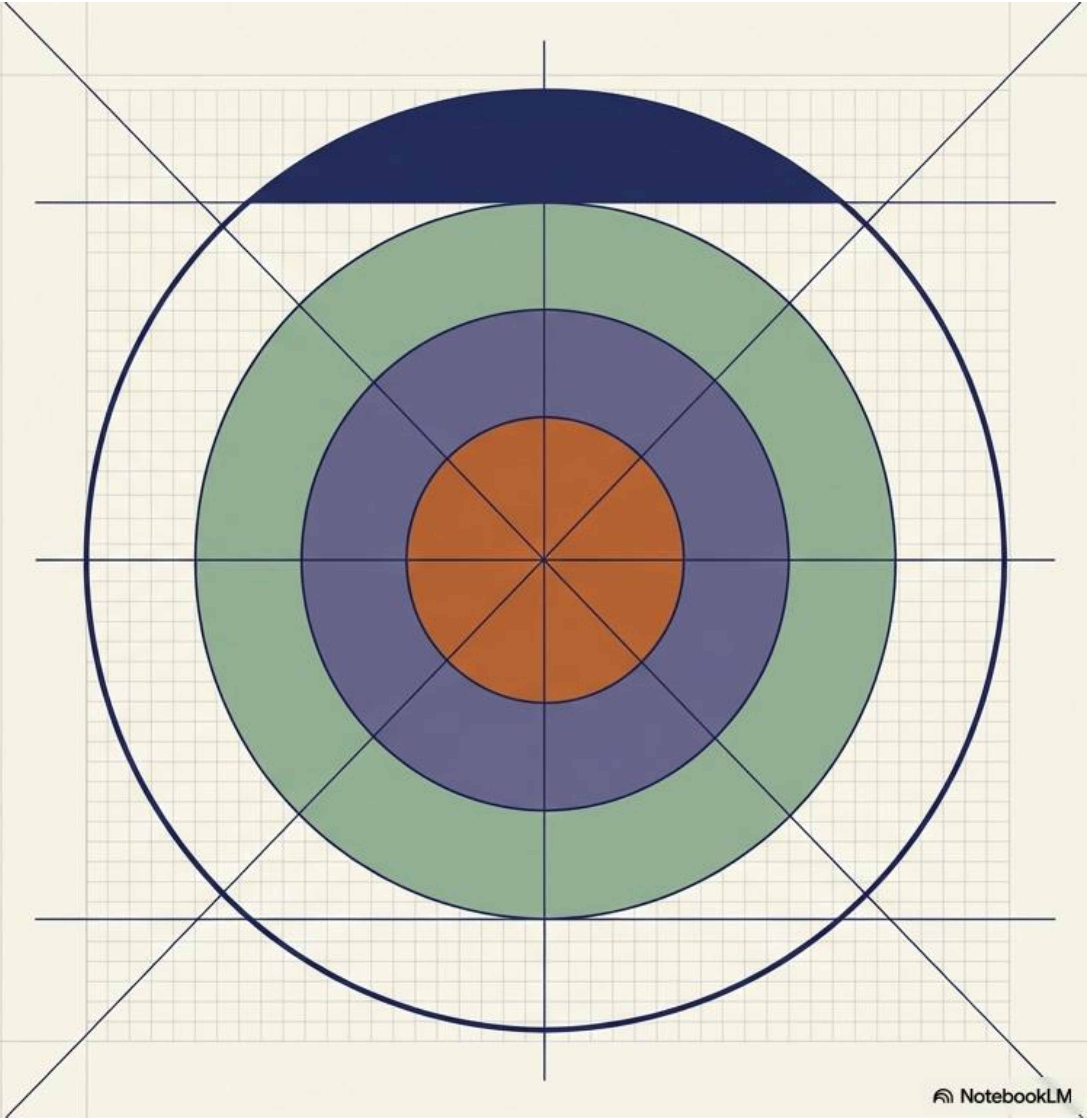
Demystifying the Artificial Intelligence Ecosystem

From Predictive Machine Learning to Generative Intelligence (Day 1)

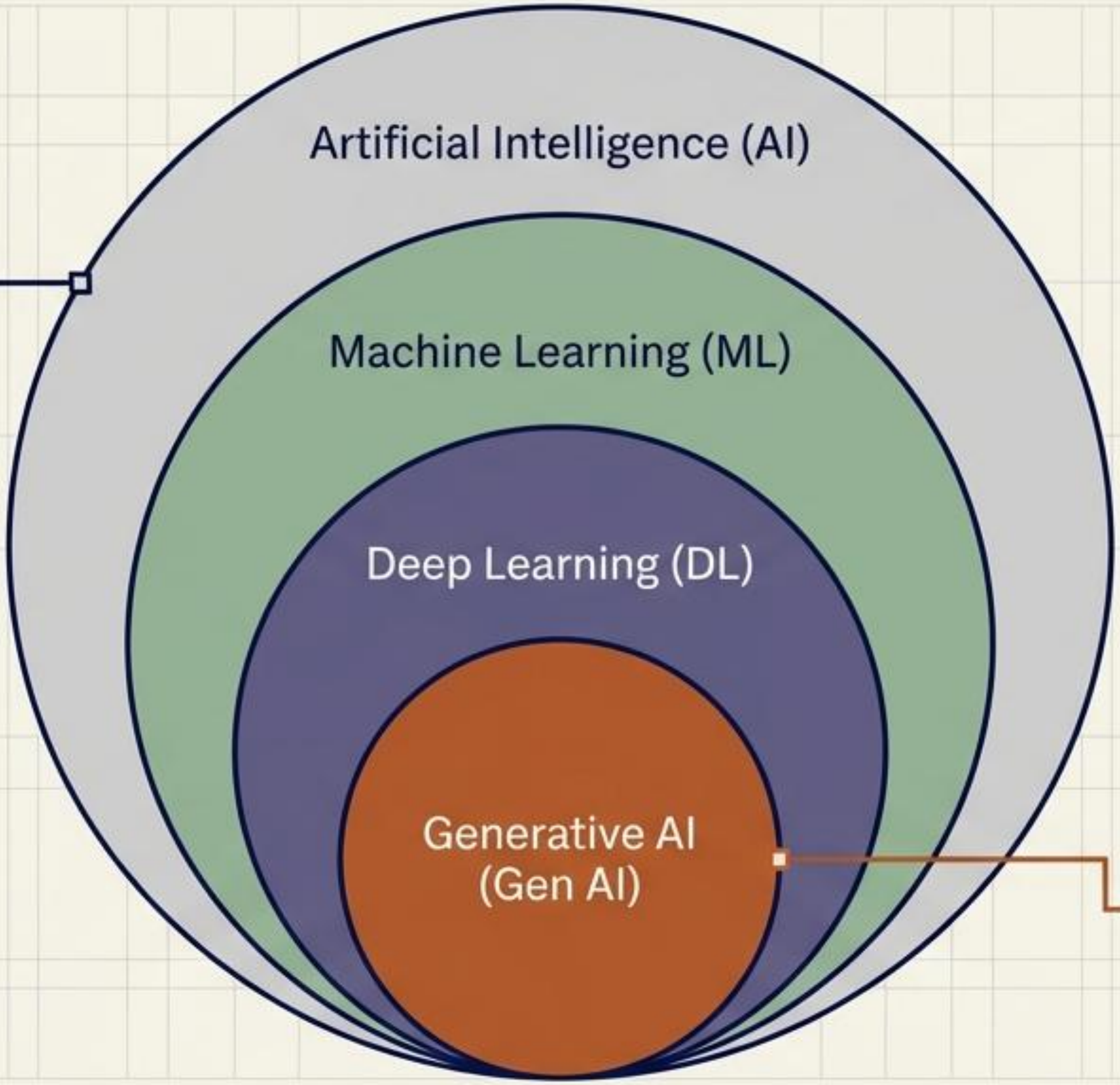
A foundational overview of data, models, and the evolution of algorithmic capabilities.



The AI landscape is a series of nested capabilities

AI Core Principle

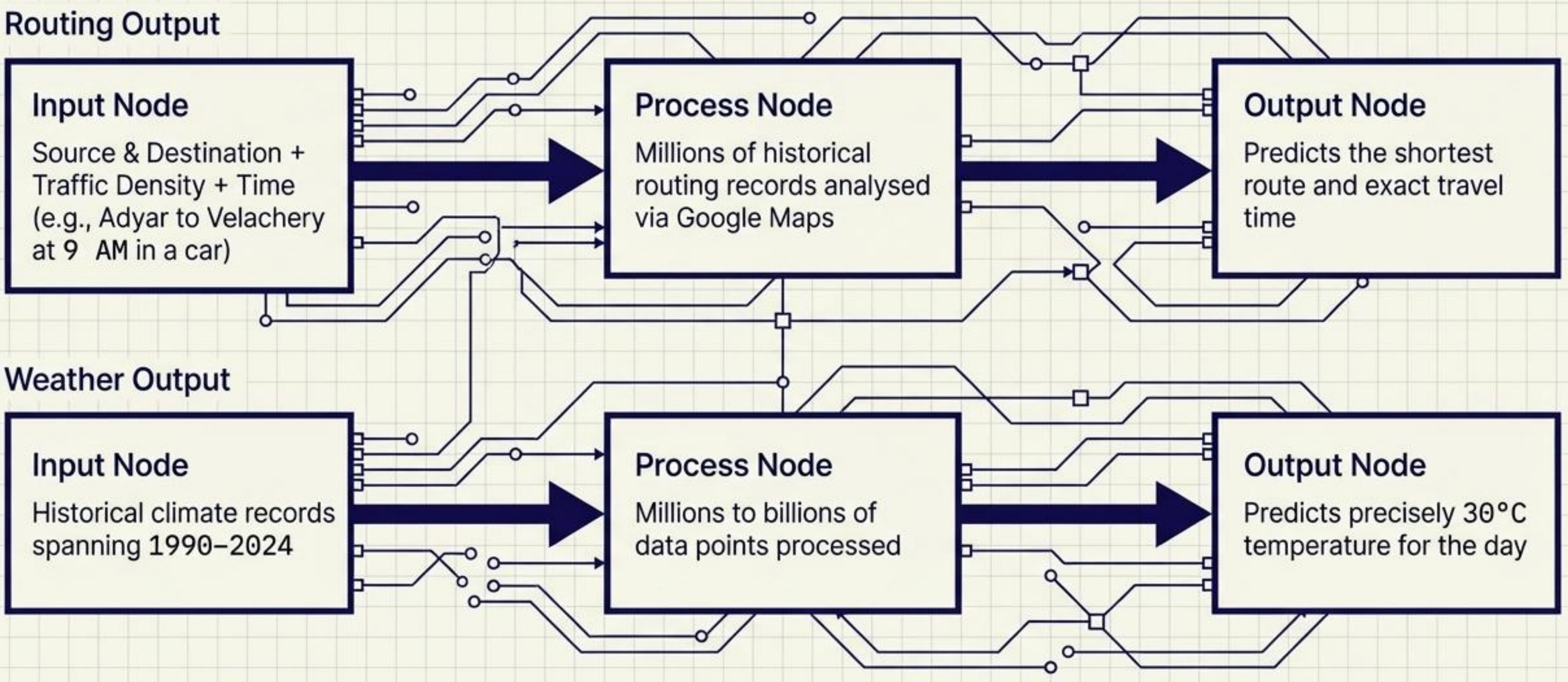
A machine mimicking human behaviour to accomplish complex tasks.



The Golden Rule

Every Generative AI is a form of Deep Learning, but not all Artificial Intelligence is Generative.

Prediction engines rely on historical data processing



SYSTEM NOTE: AI maps current variables against vast historical sets to generate high-probability forecasts.

Machine Learning automates pattern recognition

1. Data Collection
Gathering thousands to millions of records

→

2. Data Cleansing
Standardising inputs

→

3. Model Selection
Choosing the mathematical approach

→

4. Testing
Validating accuracy

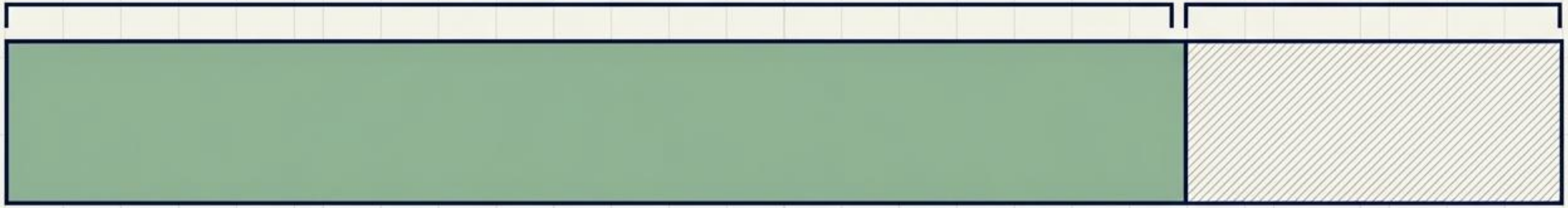
KEY TAKEAWAYS

Classification:	Scale:	Core Function: PREDICTIVE
A specific, highly structured subset of Artificial Intelligence.	Routinely processes thousands to millions of distinct records.	The definitive output of Machine Learning is a prediction based on historical patterns.

Models require strict separation of training and testing data

80%
TRAINING DATA

20%
TESTING DATA



The historical data fed into the system to teach the machine the underlying patterns and relationships.

Unseen validation data held back to verify the model's accuracy.

THE GOLDEN CONSTRAINT

The input used for testing must never be a part of the training data. If a model is tested on data it has already seen, the prediction is fundamentally invalid.

Algorithms adapt based on the presence of data labels

SUPERVISED LEARNING

Mechanism: Trained on explicitly labelled data (e.g., Content-based star ratings).

Type 1: Linear Regression

Outputs numerical values (e.g., Real estate pricing, defect prediction testing).

Type 2: Classification

Outputs specific categories (e.g., Predicting heart attack risk: Yes or No).

UNSUPERVISED LEARNING

Mechanism: Trained on unlabelled data to autonomously find hidden structures.

Application

Behavioural analysis and collaborative filtering. Groups by genre or behaviour without relying on explicit user likes or comments.

Multimodal systems process varied data streams simultaneously

Tabular Data

Age, Blood Group, Diabetic Status, Blood Pressure, Location, Phone, Genetic History, Smoker, Drinker.

Image Data

Patient Diagnostic Engine

Convolutional Neural Network (CNN)

Configured to read retinal image details. Trained on thousands to millions of historical patient records.

Categorical Classification Output

Heart Disease Risk:

YES / NO

Binary classification based on integrated analysis.

Multimodal Architecture: Advanced models accept and integrate text, numerical tables, images, and video to formulate a single predictive conclusion.

Deep Learning applies neural networks to massive datasets

Definition

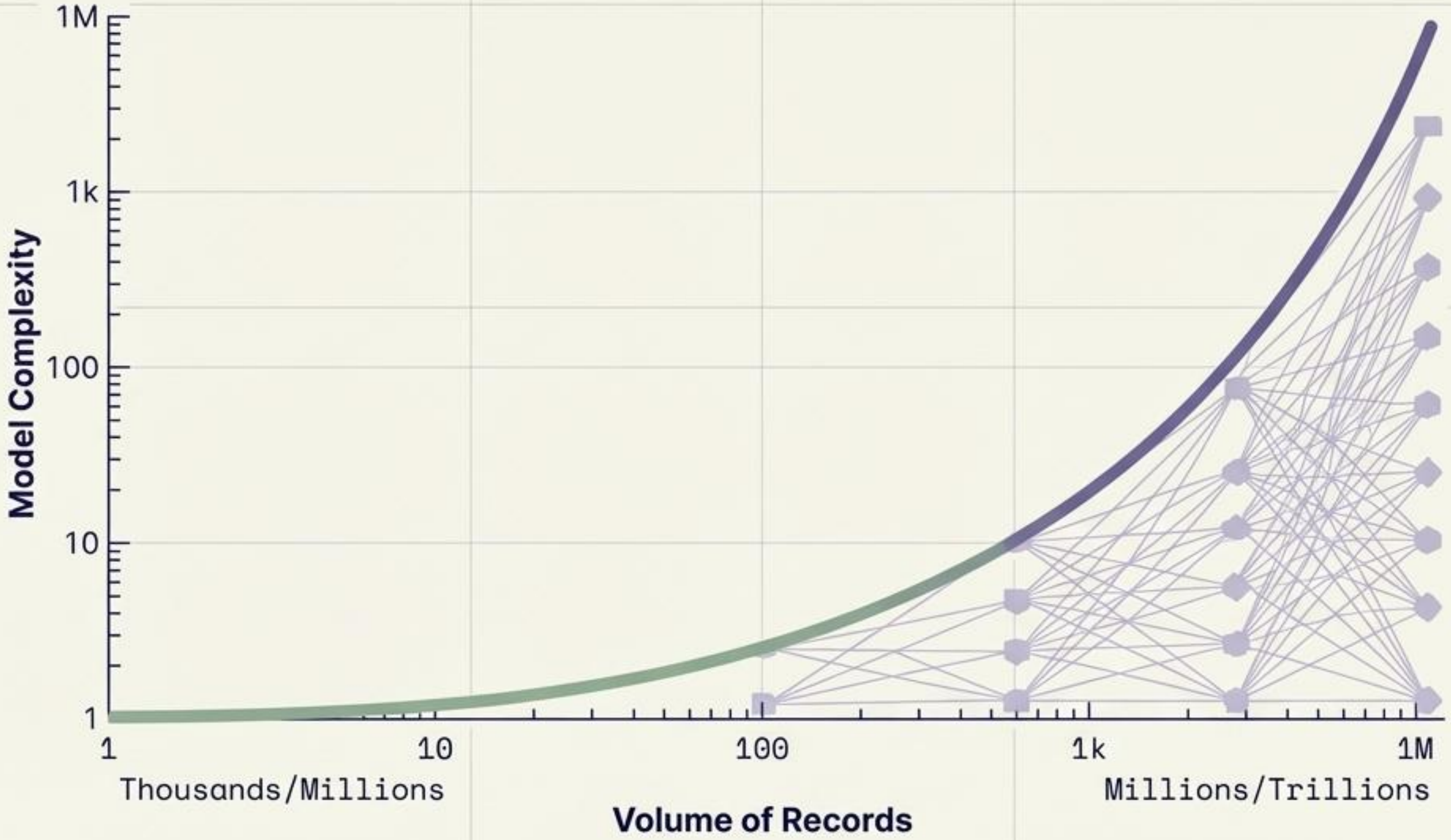
A specialised subset of Machine Learning.

The Engine

Combines the foundational predictive power of ML with complex, multi-layered Neural Networks.

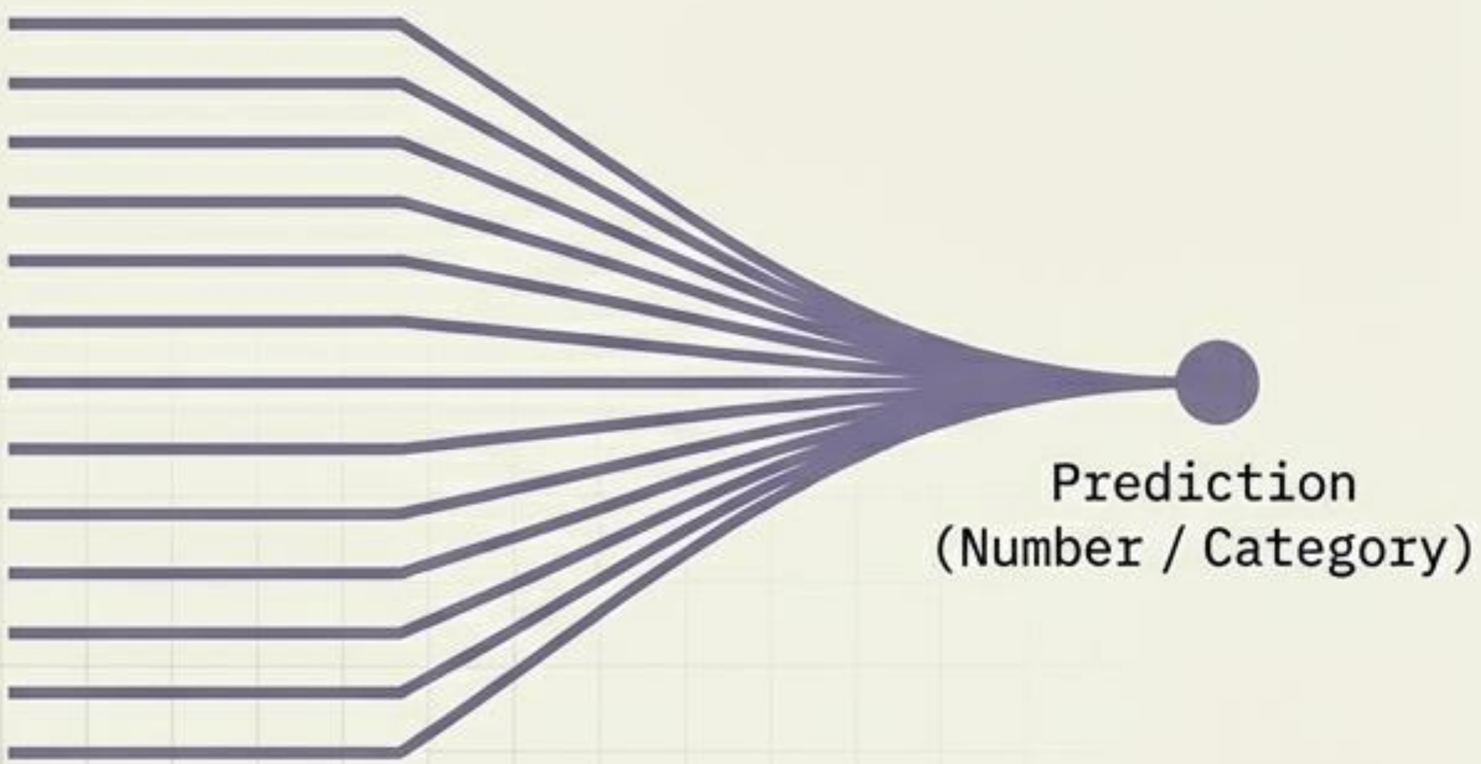
The Output

Produces highly advanced, nuanced predictive capabilities derived from vast amounts of data points.



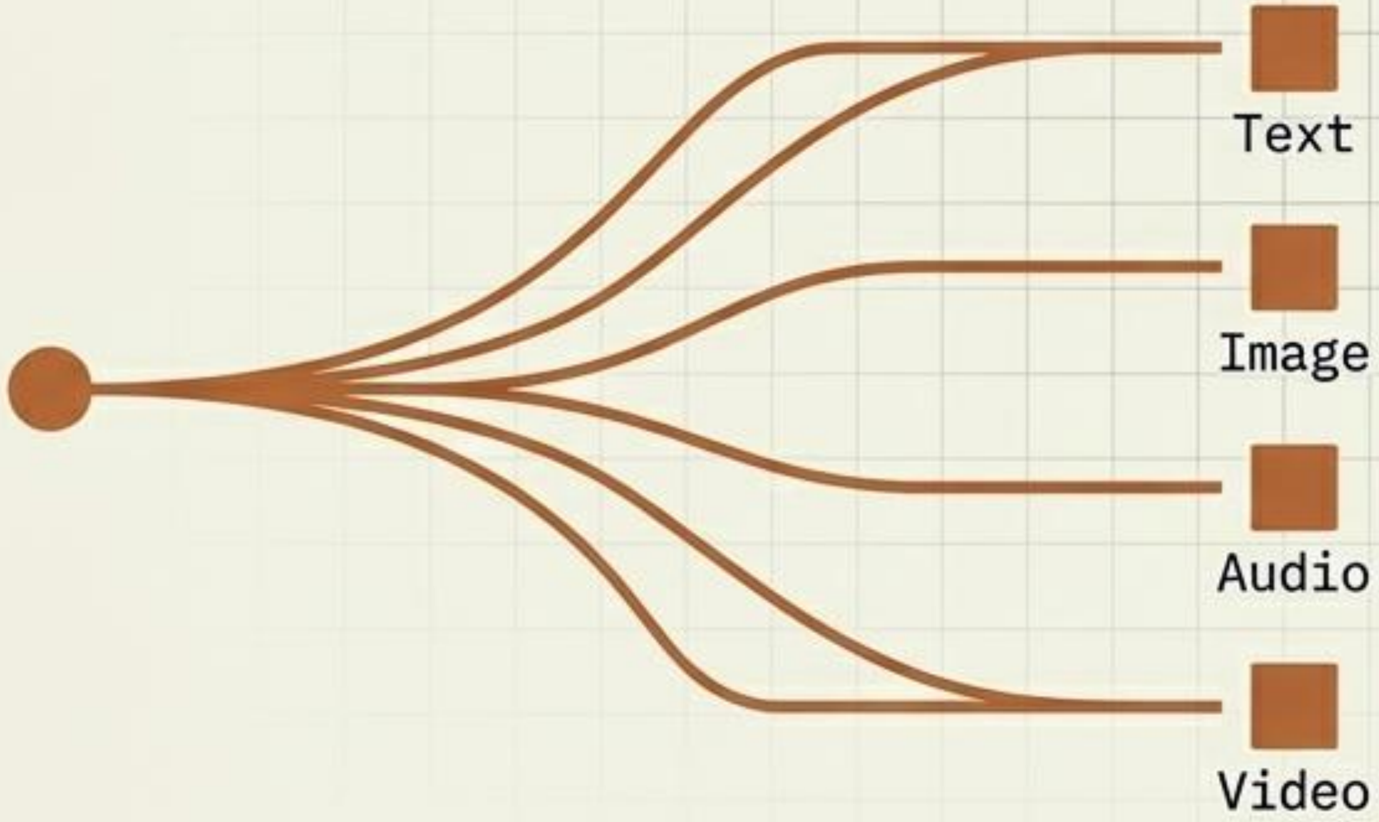
Generative AI shifts the paradigm from prediction to creation

PREDICTIVE ML/DL



Systems designed to analyse historical data and output a singular probability or classification.

GENERATIVE AI



A subset of Deep Learning. Unlike ML/DL, it is absolutely non-predictive. It generates entirely new, original content formats.

Large Language Models scale creation across all domains

Framework:

Generative Pre-trained Transformer (GPT).

Pre-trained on massive, cross-domain global datasets—not limited to software or specific fields.

Ecosystem:

Developed by providers like OpenAI; accessed via chat agents like ChatGPT.

96 to 120

Processing layers in GPT-4 architecture

190 LAYERS / 120 BILLION

Parameters actively powering GPT-5 architecture

The continuum of artificial intelligence

	Machine Learning	Deep Learning	Generative AI
Relationship	Subset of AI	Subset of ML	Subset of DL
Core Function	Predictive	Predictive	Generative
Data Volume	1,000s to Millions	Millions to Trillions	Massive Global Datasets
Mechanism	Statistical Algorithms	Neural Networks	Large Language Models (Transformers)
Output Type	Numbers or Categories	Advanced Nuanced Predictions	Original Text, Image, Audio, Code

DAY 2 PREVIEW

Next: the internal architecture of Large Language Models

Having mapped the boundaries from basic predictive routing to the generative frontier, Day 2 will deconstruct the internal workings of LLMs. Prepare to explore how Transformers process language and the precise anatomical layers driving modern generative agents.